

Naan_binary

Project Report for Phase 1 & Phase 2

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ECSCI24305 - Deep Learning: Principles and Practices

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1 Phase 1: Problem Identification & Proposal

1.1 Project Title

Ingredient-Based Food Recipe Generator with Nutritional Estimation

1.2 Problem Statement

Many users often have a limited set of ingredients available at home but do not know what dishes they can prepare from them. Existing recipe platforms require manual searching and do not provide intelligent suggestions based solely on available ingredients.

The goal of this project is to develop a deep learning-based system that:

- Takes a list of available ingredients as input
- Suggests possible dishes that can be prepared
- Generates the corresponding recipe steps
- Provides nutritional information such as calories, protein, fat, and carbohydrates

1.3 Motivation and Relevance

- Helps users reduce food waste by utilizing available ingredients
- Saves time in meal planning
- Supports healthy eating through nutritional insights
- Demonstrates a real-world application of deep learning in natural language processing and recommendation systems

1.4 Application Flow

1. User inputs available ingredients
2. Model identifies matching recipes from learned data
3. System outputs:
 - Dish name(s)
 - Recipe instructions
 - Nutritional values

1.5 Type of Data

This project will use **text-based structured data**, including:

- Recipe title
- Recipe id

- Time to prepare
- Tags for Food
- Nutritional information
- Number of steps
- Steps to prepare the food/instructions
- Food description
- Ingredients required
- Number of ingredients required
- Images

1.6 Expected Deep Learning Approach

- Text preprocessing and tokenization
- Ingredient-to-recipe mapping using:
 - Sequence models (LSTM/GRU) or Transformer-based models
- Multi-output prediction:
 - Dish classification
 - Recipe generation
 - Nutritional value prediction (regression)

2 Phase 2: Dataset Design & Collection

2.1 Dataset Source

The dataset will be created by scraping recipe data from:

Food.com

Web scraping will be performed using:

- Selenium (for dynamic content loading)
- BeautifulSoup (for HTML parsing)
- Python (data storage and processing)

This ensures compliance with the project requirement of creating an original dataset.

2.2 Dataset Structure

Field	Description
Name	Name of the dish
id	Unique identifier for each recipe
minutes	Time to prepare
tags	Type/Category of dish
Nutrition	Nutritional values
n_steps	Number of steps in the recipe
steps	Steps in the recipe
Description	Description of the recipe
Ingredients	Ingredients required
n_ingredients	Number of ingredients
image_url	Images

2.3 Data Format

The dataset will be stored in:

- CSV format for structured storage
- JSON format for hierarchical ingredient and instruction data

Example structure:

```
{
  "name": "Vegetable Pasta",
  "id": 1,
  "minutes": 60,
  "tags": [italian,easy to prepare],
  "nutrition":["calories": 320,"protein": 12],
  "n_steps":5,
  "steps": ["..."],
```

```
"Description": a healthy and quick to make pasta
"ingredients": ["pasta", "tomato", "onion"],
"n_ingredients":3,
"images":"image_url"
}
```

2.4 Dataset Size

- Target size: 20,000+ recipes (initial phase)
- Final goal: 100,000+ recipes (if time permits)

2.5 Data Collection Process

1. Get links: read sitemap index and sitemap .xml.gz files.
2. Filter URLs: keep only valid /recipe/...-id links.
3. Scrape pages: fetch each URL and parse recipe fields (title, time, ingredients, steps, nutrition).
4. Save progress: write CSV checkpoint every 100 recipes.
5. Finalize data: save food.com.2k.csv and quickly verify rows/columns.
6. Store extracted data into CSVJSON
7. Maintain logs of:
 - Number of pages scraped
 - Number of recipes collected
 - Errors or missing data

2.6 Data Collection Logs and Monitoring

During the scraping process, logs are maintained to monitor the progress and ensure data quality. The logs include:

- Number of recipes scraped over time
- Status of each scraping batch
- Failed URLs or parsing errors
- Dataset growth statistics

These logs help in debugging issues, tracking dataset size, and ensuring reliable data collection.

```

Tarun D:\..\DL_Food 9.295s c:/python313/python.exe test.py
Starting scrape for 2000 recipes...
Collected 2000/2000 recipe URLs
Scraped (1/2000): Gravy With Sausage or Hamburger
Scraped (2/2000): Grilled Herbed Shrimp on Mixed Greens
Scraped (3/2000): Music Parents Famous Frito Pie
Scraped (4/2000): Shish Kabob Marinade
Scraped (5/2000): Orange Cordial Fudge Balls
Scraped (6/2000): Pan-fried Potatoes
Scraped (7/2000): Cream Cheese Icing
Scraped (8/2000): Baked Apple French Toast
Scraped (9/2000): Apple Crumble
Scraped (10/2000): All to Myself Cheesecake
Scraped (50/2000): Taco Baked Potatoes
Scraped (100/2000): Braised Pork (Tenderloin) With Mushrooms
Checkpoint saved: 100 rows -> food_com_first2k.csv
Scraped (150/2000): Skillet Enchiladas
Scraped (200/2000): Skor Bar Sugar Cookies
Checkpoint saved: 200 rows -> food_com_first2k.csv
Scraped (250/2000): Kansas Dirt Cake
Scraped (300/2000): Potatoes With Onion Sauce and Parmesan Cheese
Checkpoint saved: 300 rows -> food_com_first2k.csv
Scraped (350/2000): Pear and Ginger Crisp With Spiced Whipped Cream
Scraped (400/2000): Chocolate Tray Bake
Checkpoint saved: 400 rows -> food_com_first2k.csv
Scraped (450/2000): Ancho Chile and Citrus Ketchup
Scraped (500/2000): Toffee Brownies
Checkpoint saved: 500 rows -> food_com_first2k.csv

```

Figure 1: Scraping Progress Log Showing Number of Recipes Collected

```

Checkpoint saved: 1400 rows -> food_com_first2k.csv
Scraped (1450/2000): Lasagna With Saucy Sausage, Peppers & Onions
Scraped (1500/2000): Stuffed Mushrooms
Checkpoint saved: 1500 rows -> food_com_first2k.csv
Scraped (1550/2000): Dry Jell-O Salad (Jello) Cottage Cheese Fruit Salad
Scraped (1600/2000): Broccoli Supreme
Checkpoint saved: 1600 rows -> food_com_first2k.csv
Scraped (1650/2000): Blueberry and Blackberry Pie
Scraped (1700/2000): Lemon Drop
Checkpoint saved: 1700 rows -> food_com_first2k.csv
Scraped (1750/2000): Broiled Tomato Slices With Gouda Cheese
Scraped (1800/2000): Chocolate Topped Apple Cake
Checkpoint saved: 1800 rows -> food_com_first2k.csv
Scraped (1850/2000): Evelyn's Pickled Okra
Scraped (1900/2000): Focaccia Rounds
Checkpoint saved: 1900 rows -> food_com_first2k.csv
Scraped (1950/2000): Banana Cake With Caramel Frosting
Scraped (2000/2000): Honey-lacquered Duck With Sour Cherry Sauce
Checkpoint saved: 2000 rows -> food_com_first2k.csv
Scraping complete. File saved as 'food_com_first2k.csv' with 2000 rows

```

Figure 2: Successfully scraped 2000 rows

2.7 Data Quality Considerations

- Remove duplicate recipes
- Handle missing nutritional values
- Standardize ingredient names
- Clean text by removing special characters and HTML tags

2.8 Ethical Considerations

- Data is collected only for academic purposes
- No copyrighted content will be redistributed commercially
- Website scraping will follow polite crawling practices (rate limiting)